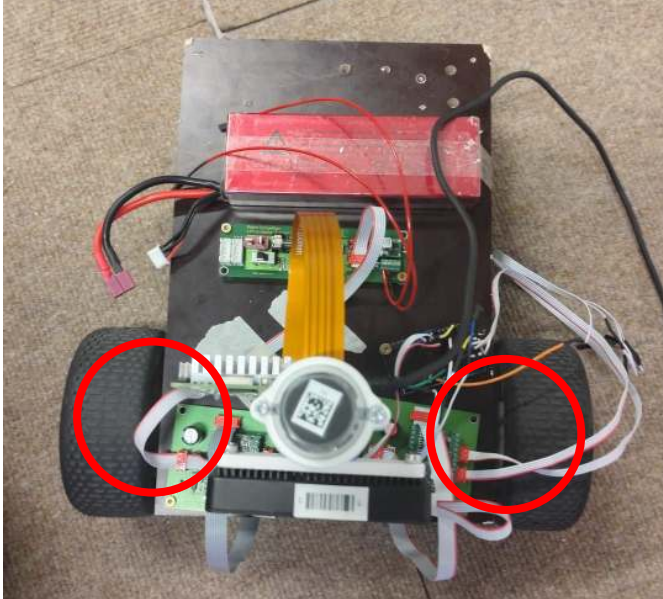


Focus on **Motion** Not on electronics

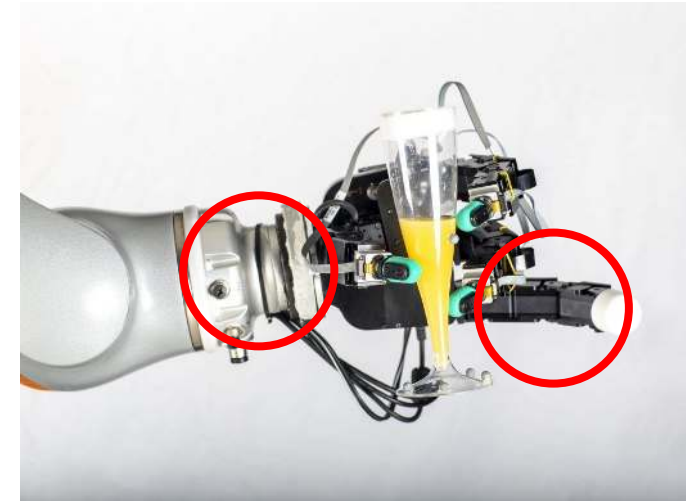
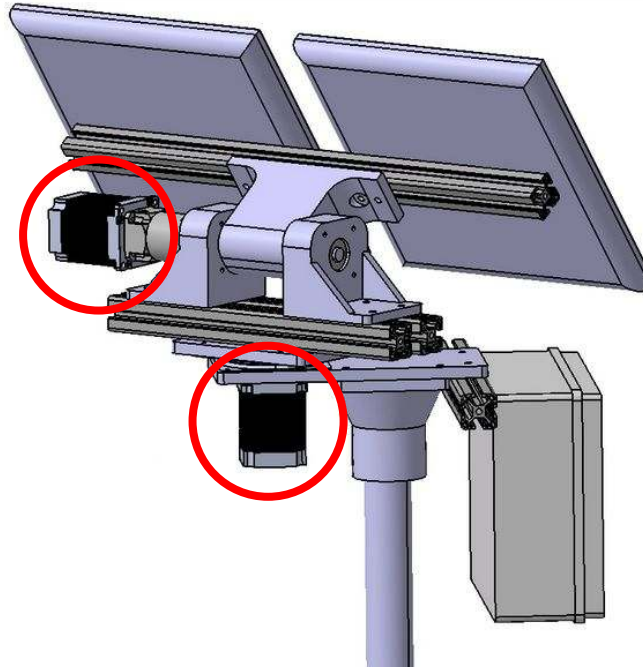
Democratized Advanced Actuation

Problem

Making Things Move is a Nightmare



Student projects

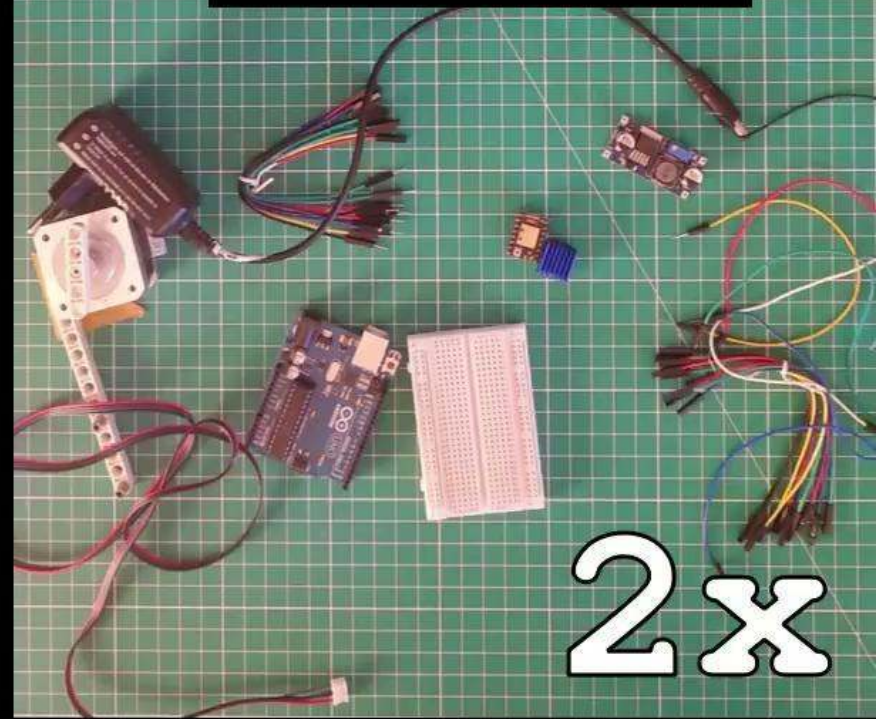


LASA Lab, EPFL

Slow

Difficult

Expensive

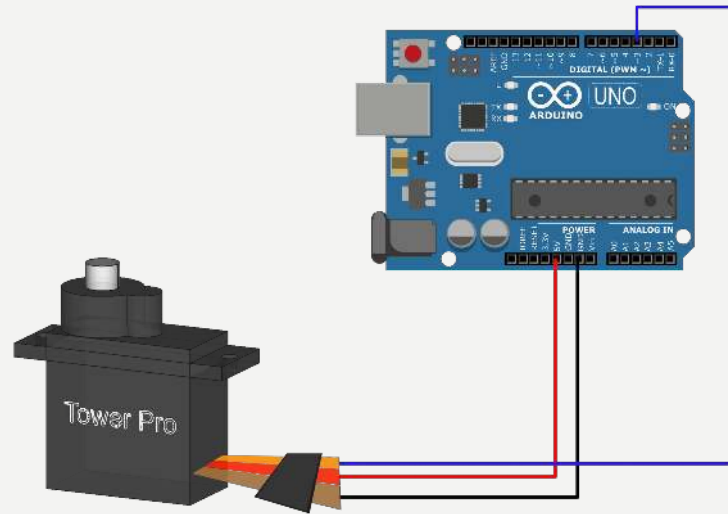


Problem

Making Things Move is a Nightmare

+10x time lost

on even the basic setup



+25% expenses

on professional products wasted

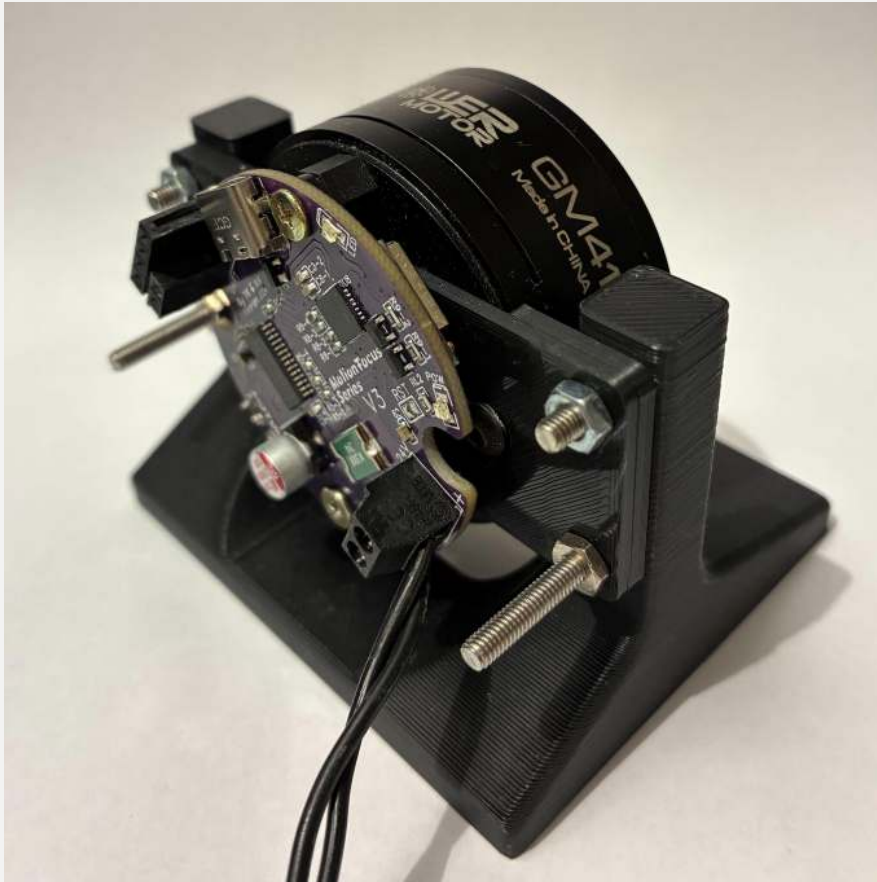


Slow Innovation : Because the setup is so painful, bringing an idea to reality takes forever.

We accelerate **prototyping**
And make it accessible to **everyone**

Solution & MVP – Hardware Kit

Plug-and-Play Actuator for Rapid Prototyping



↻ Global Integration

Versatile and fail-safe

🔌 Intelligent Control

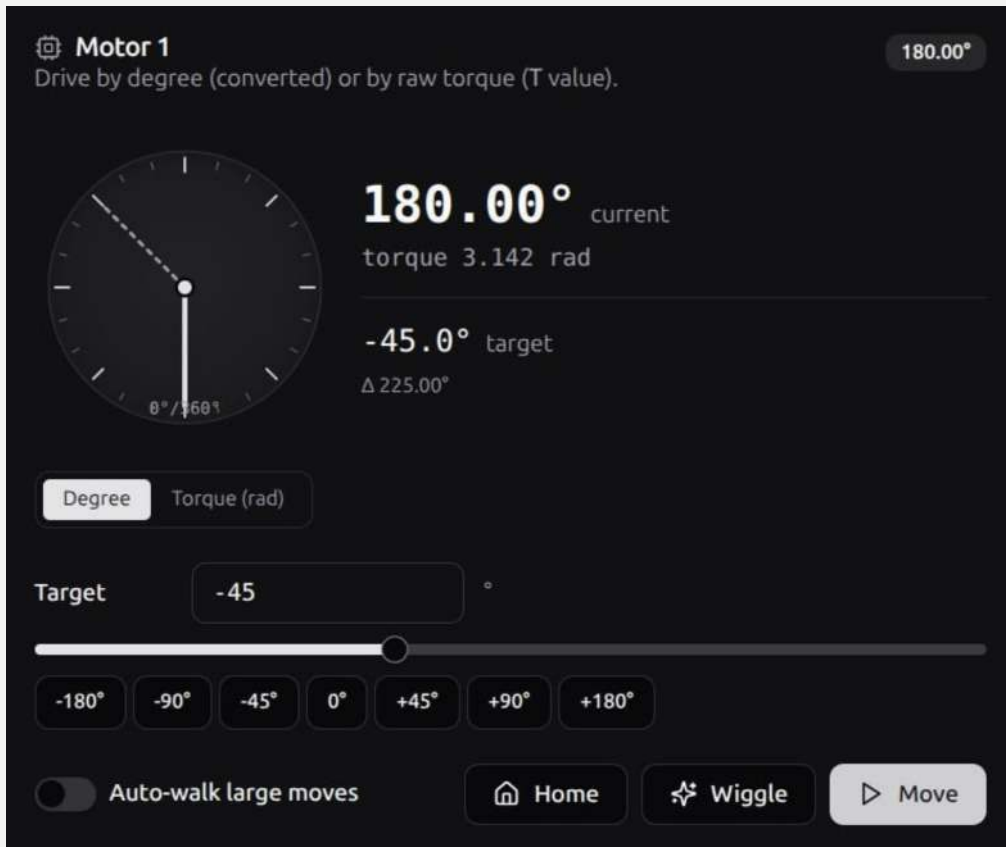
High Performance and control

\$ Affordable

Hardware around 150 CHF

Solution & MVP – Software

Plug-and-Play Actuator for Rapid Prototyping



Ultra Low Learning Curve

Intuitive and easy to use software

AI agent, Python SDK, Web UI

Collaborative

Work on the same hardware than your friends

Open source

Free for everyone to use and modify

Proven Validation from Real Users

"We would buy this if it existed" – A Robopoly Association member, EPFL



Inbound Interest

20+ people asked to test our product



Associations leaving competitors

EPFL Rocket Team left Maxon for DIY Arduino over complexity/price



Same bottlenecks

Robotics clubs, student associations, competitions, DIY tinkerers



Paying for less

Teachers buy 100+ CHF Thymios to flip them and just use their wheel

Competition

Dynamixel/Maxon

Industrial but expensive & proprietary



Motion Focus

Professional and DIY simplicity



Arduino

Cheap but complex and limited



Our edge: affordable + compatible + simple.

We democratize **Motors** like
Arduino did with Microcontrollers.

Market

🎯 Beachhead

Tech students, teachers & makers



Become credible locally → build up appeal for the broader market → expand globally

zhaw School of Engineering

ETH zürich

Les ATELIERS
de Renens

Hes·so

EPFL

How did Arduino / Raspberry Succeed ?

Abstraction

Abstracting the **complexity** of microcontrollers.

We abstract the **electronic setup** of high-performance motors.

Community Standards

Creating a **standard library** and plug-and-play format.

Easy adoption for students, carry it into their careers.

Rapid Prototyping

Reducing time-to-first-motion from **hours to minutes**.

Friction reduction is the primary driver of adoption and loyalty.

Our Business Model

We teach the use motors for the new generation of engineers

Open source Software

Free

Customer lock in through ultra intuitive
and easy to use software. **Zero friction**



Hardware Kit

150.- CHF / Motor

Total cost: ~100 CHF

Profit/kit: 50 CHF

Monetization through educational kits and accessories

Financing

Q3



2027



Crowdfund + grants

Crowdfunding
(WeMakelt), TalentKick,
Venture, EPFL support,
bootstrap

Angel seed round

Reinvest profits + loan
(FIT grant) + angel round

Motion Focus is Not Just a
Product;
It Is a **Market Entry**

Growth Plan

How we will scale

Now – 6 mo

01

Open-Source Seed

Github hardware & firmware ship

Seed the maker community in CH

→ At least 500 units

6 – 18 mo

02

Establish with makers

100+ makerspaces in CH & EU.

Open-source virality.

→ 100 units / makerspace / year

18 mo+

03

World Wide solution

Transition to the industry

Engineers familiar with solution

→ Become the motion standard

The Team



Tim Kobler
Robotic Engineer



Jan Juraszek
Electrical Engineer



Veronika Wannack
Bioelectrical engineer



Nabil Bellamy
Microtechnical engineer



Florian Déjean
Data Scientist



Empowering the Next Gen of Makers

Apr–May



Software MVP

Successfully run user trials with 10 beta testers

May–Aug



First Pilots

Deploy to 2 maker-spaces and one teacher

Sep+



First Revenue

Begin outreach to farther makerspaces

Order NOW !

Focus on **Motion**
Not on electronics

Appendices

Funding

First 6 months

We raise 25k (bootstrapped)

- + Grants (Venture Kick = 40k, TalentKick = 5k, Blaze = 10k) fuel **marketing** and **more motors** = **55k**
- + **Crowdfunding** = **20k**

=100k raised /6 months

Target: approx 500 units sold (50k investment) + marketing/sales + extra (50k)

Cofounder salary = 0 CHF

6-18 months

To break even:

7 full time*5000/month = 35000.- month/salary

- + 700 units sold /month = (7000 CHF) for production per month

=42000 CHF/month

= 504'000 CHF/year

25k profit + 100k FIT grant = 125k in bank

= 379'000 CHF

In 6 months:

Target → 400k angel investment in 6 months

Viability calculations

Hardware

Students/makers

Price: 150 CHF, cost 100 CHF

Profit of 50 CHF/kit

Cost breakdown

- motor (model dependent) : 50.-
- electronics : 20.-
- assembly and mechanics: 10.-
- delivery : 9.-
- packaging: 5-
- payment fee : 1.-
- ADD-ONS : 5.-

Markup: $(\text{price} - \text{cost}) / \text{cost} = 50\%$

Margin: $(\text{price} - \text{cost}) / \text{price} = 33\%$

Software

Software: FREE for all users

- Drives adoption, zero friction

Per maker client (unit economics)

Revenue: 260.- (2 hardware kits)

We estimate ~2 kits/year per client

Cost: 246.-

200.- (2 hardware units)

$12 \times 3.- = 36.-$ (SW cost/yr)

CAC: 10.- (rough)

$(\text{Price} - \text{cost}) / \text{cost} = 0.06$ (approx break even)

Estimated Market Size

TAM 100M Emerging robotics companies

SAM 15M EU Robotics Startup

SOM 50K CH Prototyping

Market size calculations

TAM: EU technical SMBs

10B is the market size of electrical actuators in Robotic companies

2B are integrated actuators

10% of this 2B are Emerging Robotic Companies

50% of this companies need a faster market entry

TAM: 100M CHF

SAM = Swiss technical SMBs

15% of the TAM is european

SAM: 15M CHF

SOM Swiss technical schools, labs

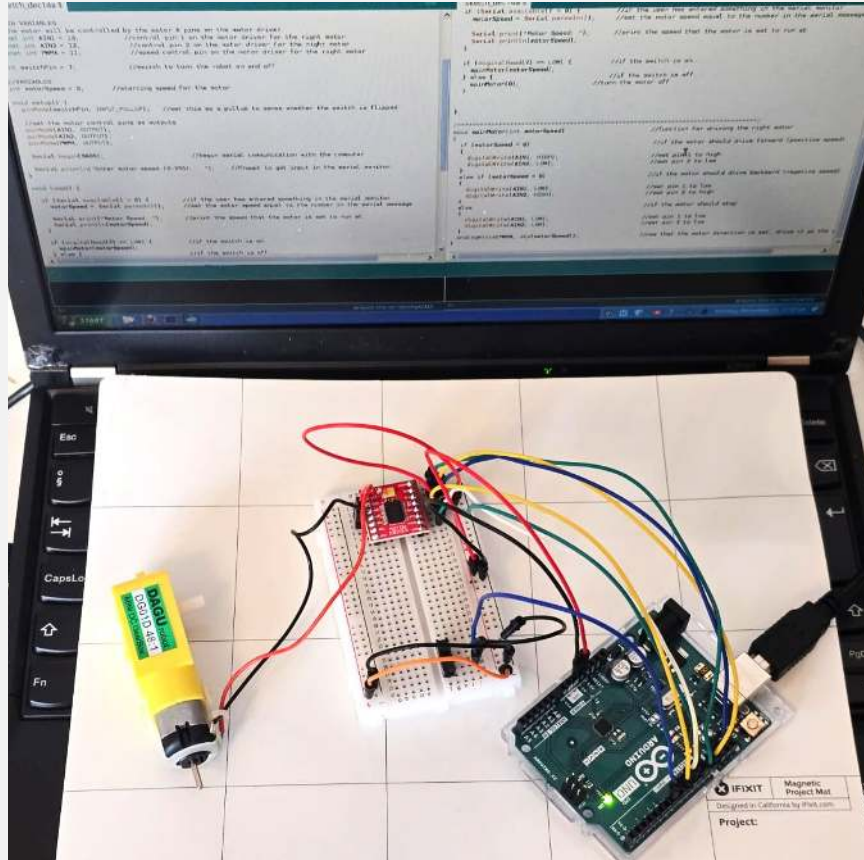
More than **1000** active Makers in Switzerland

20% would use a plug and play solution

100 clients with a 500.- CHF budget

SOM: 50'000.- CHF

Validation tests



Real setup for arduino motor guide

```
// BLDCMotor motor = BLDCMotor(pole pair number);
BLDCMotor motor = BLDCMotor(MotionFocus::POLE_PAIR);
// BLDCDriver3PWM driver = BLDCDriver3PWM(pwmA, pwmB, pwmC, Enable(optional));
BLDCDriver3PWM driver = BLDCDriver3PWM(MotionFocus::PHASE_U,
                                         MotionFocus::PHASE_V,
                                         MotionFocus::PHASE_W,
                                         MotionFocus::DRIVER_ON);

// angle set point variable
float target_angle = 100;

// instantiate the commander
Commander command = Commander(Serial);
void doTarget(char* cmd) { command.scalar(&target_angle, cmd); }

void setup() {
    // initialize LED pin
    pinMode(MotionFocus::LED, OUTPUT);
    //strating init phase
    digitalWrite(MotionFocus::LED, HIGH);

    // initialise magnetic sensor hardware
    sensor.init();
    // link the motor to the sensor
    motor.linkSensor(&sensor);

    // driver config
    // power supply voltage [V]
    driver.voltage_power_supply = 12;
    driver.init();
    // link the motor and the driver
    motor.linkDriver(&driver);
}
```

BLDC drive tests

Real-world cases we witnessed:

Rocket team GS division project:

- They initially considered maxon motors
- Too expensive and difficult to find the correct one and also stick with it
- They opted for buying their own motor, drivers, and Arduino separately
 - This took time to figure out
 - Good for prototyping
 - Yet, they need a robust, soldered and PCB version later, meaning they need to do extra work and discard the Arduino
- Our solution needs no figuring out, can be used for prototyping and for final designs

Some (Arduino) motor workshops

- The controllers, extra boards and shields, and the motors all need to be setup, tested
- If it is for the sake of simply teaching motors, this adds unnecessary complexity and time
- Ours has one app install and one connection
- (Attended a motor workshop, only went through theory because an interactive setup would be too time consuming)

Competitor tests

Motor setup timings:

Simple test: Using bare components you can buy, connect it and make the motor rotate by 90°

Setup	Motion focus	Arduino + servo motor	Arduino + DC motor	Arduino + stepper motor
Time	~1 min	~10 min	~30 min	~ 60 min (not finished)

→ We can safely say that we speed up at least 10x compared to traditional methods

Motor setup costs:

Setup	Motion focus	Arduino + stepper motor	Maxon motor + components (cheapest configuration)
Cost (CHF)	~150	~50	~200

→ We can safely say that we are at least 25% cheaper

Beachhead selection

Meeting with high-school teachers

- The highlight for them is that our idea removes the “brain” part and its standalone
- One pointed out they like it if it could be miniature to save space on intricate projects
- Making very simple projects becomes more fun (as they get brought to life) with the motors
- They said Vex robotics isn’t exactly a direct competitor
 - they are focused on intricate robot building for competitions, and are their own ecosystem and are incompatible with legos and other parts (which is how they make money)
 - The way we stand out is that ours can be used with anything (legos, screws, etc) and are free of any restrictions
- Thymio would be the closest competitor
 - apparently lots of thymio projects were using the thymio laying on its side just to use its wheels to move things – we are exactly that.
 - Could be used for any kind of project where you want to make something simple
 - the teacher said they have some robot (animatronics) that they 3D print and without any motor solution like ours it’s not feasible to make them move
 - Arduino is also close, but not used as much in education
- They think the DIY market is actually bigger than we thought
 - could be used for:
 - Robot competitions
 - Home DIY
 - RC hobbyists
 - Underwater vehicles
 - Watches/clocks

Conclusion: While the education niche is feasible, they clearly preferred the DIY approach, hence DIY as a starting point for popularization is a plus.

We spoke with two high school teachers and two lab assistants.

It appears to tailor our solution to education or labs, it would require lots of pre-development and both parties strongly mentioned DIY (without us mentioning it) – leading to the conclusion that DIY is the easiest, and the fastest solution for us to grow before we are able to enter the larger market.

Meeting with lab assistants

No sense for lab that uses hardware a lot (they use their own mcus, motors)

Dynamixel already exists but:

- they are expensive (especially the bigger ones)
- you can’t program the microcontroller directly, only through buses
- Still need expertise on hardware and motors (but not a problem in the case of this lab for example)

He said it would be more useful for other labs that don’t touch hardware at all, like chemistry ones who just want to move something. to keep in mind;

- it would need to work on different OSs
- library interfaces with different environments such as MATLAB

Conclusion: Seems like quite some work for the lab approach to make it usable for any scenario. It appears that the main hindrance is that labs in engineering are already experienced enough anyways to use our competitors. Labs that do not have such knowledge could even have difficulty using our motor mechanically, like mounting it to a system or creating such a system from scratch. This suggests that our idea would only apply to labs if we offer a seamless service where they don’t have to worry about the mechanical part either – something we will need to develop.